

Disproof of conjecture 00000000429

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Verdict: FALSE

We disprove conjecture 00000000429 as stated. The refutation is unconditional, completely elementary, and rests on a single explicit value:

$$\begin{bmatrix} 6 \\ 2 \end{bmatrix}_{q=-1} = 3,$$

and 3 is neither 2^t nor -2^t for any integer t . Hence *no* formula $v(n, k)$ whatsoever can make the value of $\begin{bmatrix} n \\ k \end{bmatrix}_q$ at $q = -1$ equal to $2^{v(n, k)}$ for all n, k , and the same witness defeats the weaker “signed” reading value $= \pm 2^{v(n, k)}$.

1 The conjecture, quoted verbatim

From `conjectures/00000000429.md`:

English. Definition: The q -binomial $[n \text{ choose } k]_q$ at $q = 1$ counts binomial coefficients.

Conjecture: Its value at $q = -1$ is of the explicit form $2^{v(n, k)}$ (v a 2-adic valuation formula, giving a complete signed version).

The conjecture is filed in both English and Chinese. The Chinese original is reproduced verbatim (in Unicode) in `README.md`; the PDF font here does not embed CJK glyphs, so the Chinese is kept in the markdown copy rather than duplicated. The two versions are identical in content.

2 How the statement must be read

The filed text is informal in one essential respect, and we record explicitly which reading we refute.

1. **The claim is about the value, not only the valuation.** The sentence says the value *is of the explicit form* $2^{v(n, k)}$. Taken literally this asserts that every value $\begin{bmatrix} n \\ k \end{bmatrix}_{q=-1}$ is a power of two. This is the primary reading, and it is false.
2. **The parenthetical invites a weaker reading.** “Giving a complete signed version” can be read as allowing a sign, i.e. $\begin{bmatrix} n \\ k \end{bmatrix}_{q=-1} = \pm 2^{v(n, k)}$. This weaker reading is *also* false, by the same witness ($3 \neq \pm 2^t$ for any t).
3. **v is a nonnegative integer.** A “2-adic valuation formula” produces an exponent $v(n, k) \in \mathbb{N}$; thus $2^{v(n, k)}$ is a nonnegative integer power of two, and $\pm 2^{v(n, k)}$ is a *signed* power of two (with $\pm 2^0 = \pm 1$). The witness 3 lies outside both families.

4. **Domain.** $0 \leq k \leq n$, as usual for q-binomial coefficients; $\begin{bmatrix} n \\ k \end{bmatrix}_{q=-1} = 0$ for $k > n$.

3 The value at $q = -1$: recursion and table

Definition 1 (q-binomial). For $0 \leq k \leq n$,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(1 - q^n)(1 - q^{n-1}) \cdots (1 - q^{n-k+1})}{(1 - q)(1 - q^2) \cdots (1 - q^k)}$$

is a polynomial in q with nonnegative integer coefficients (the Gaussian binomial coefficient). Equivalently it is determined by the Pascal recursion

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q + q^k \begin{bmatrix} n-1 \\ k \end{bmatrix}_q, \quad \begin{bmatrix} n \\ 0 \end{bmatrix}_q = 1, \quad \begin{bmatrix} 0 \\ k \end{bmatrix}_q = 0 \ (k \geq 1).$$

The product formula is 0/0 at $q = -1$ and must not be substituted directly; the value $\begin{bmatrix} n \\ k \end{bmatrix}_{q=-1}$ is obtained from the exact polynomial. Setting $q = -1$ in the recursion gives the *integer* recursion

$$\begin{bmatrix} n \\ k \end{bmatrix}_{q=-1} = \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_{q=-1} + (-1)^k \begin{bmatrix} n-1 \\ k \end{bmatrix}_{q=-1}, \quad \begin{bmatrix} n \\ 0 \end{bmatrix}_{q=-1} = 1, \quad \begin{bmatrix} 0 \\ k \end{bmatrix}_{q=-1} = 0 \ (k \geq 1), \quad (1)$$

which is what both `reproduce.py` (for $n \leq 20$, cross-checked against the exact polynomial for $n \leq 12$) and the Lean development use.

The table for $n \leq 12$ (the entry in row n , column k is $\begin{bmatrix} n \\ k \end{bmatrix}_{q=-1}$; blanks are out-of-range $k > n$) is:

$n \backslash k$	0	1	2	3	4	5	6	7	8	9	10	11	12
0	1												
1	1	1											
2	1	0	1										
3	1	1	1	1									
4	1	0	2	0	1								
5	1	1	2	2	1	1							
6	1	0	3	0	3	0	1						
7	1	1	3	3	3	3	1	1					
8	1	0	4	0	6	0	4	0	1				
9	1	1	4	4	6	6	4	4	1	1			
10	1	0	5	0	10	0	10	0	5	0	1		
11	1	1	5	5	10	10	10	10	5	5	1	1	
12	1	0	6	0	15	0	20	0	15	0	6	0	1

For instance, from the definition,

$$\begin{bmatrix} 6 \\ 2 \end{bmatrix}_q = 1 + q + 2q^2 + 2q^3 + 3q^4 + 2q^5 + 2q^6 + q^7 + q^8,$$

so substituting $q = -1$ gives $1 - 1 + 2 - 2 + 3 - 2 + 2 - 1 + 1 = 3$. Similarly

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix}_q = 1 + q \Rightarrow \begin{bmatrix} 2 \\ 1 \end{bmatrix}_{q=-1} = 0, \quad \begin{bmatrix} 8 \\ 4 \end{bmatrix}_q (-1) = 6, \quad \begin{bmatrix} 10 \\ 2 \end{bmatrix}_q (-1) = 5, \quad \begin{bmatrix} 7 \\ 3 \end{bmatrix}_q (-1) = 3.$$

4 The refutation

Theorem 2 (Main). *Let $v : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ be arbitrary. Then it is not the case that*

$$\begin{bmatrix} n \\ k \end{bmatrix}_{q=-1} = 2^{v(n,k)} \quad \text{for all } n, k \in \mathbb{N}.$$

Consequently the conjecture is false as filed, for every candidate valuation formula v , and not merely for one choice of v .

Proof. Evaluate at $(n, k) = (6, 2)$. The recursion (1) gives the table entry $\begin{bmatrix} 6 \\ 2 \end{bmatrix}_{q=-1} = 3$ (verified independently by the exact polynomial $1 + q + 2q^2 + 2q^3 + 3q^4 + 2q^5 + 2q^6 + q^7 + q^8$ at $q = -1$, and formally in Lean by `decide`). Now $2^0 = 1$, $2^1 = 2$, $2^2 = 4$, and 2^t is strictly increasing in t for $t \geq 0$, so

$$2^t = 3 \implies 2^1 < 2^t < 2^2 \implies 1 < t < 2,$$

impossible for $t \in \mathbb{N}$. Hence $2^{v(6,2)} \neq 3 = \begin{bmatrix} 6 \\ 2 \end{bmatrix}_{q=-1}$ for every $v(6, 2)$, so no formula v works at $(n, k) = (6, 2)$. $\square$

Corollary 3 (Signed reading). *Even the weaker reading $\begin{bmatrix} n \\ k \end{bmatrix}_{q=-1} = \pm 2^{v(n,k)}$ is false: for every $t \in \mathbb{Z}$, $2^t \neq 3$ and $-2^t \neq 3$, while $\begin{bmatrix} 6 \\ 2 \end{bmatrix}_{q=-1} = 3$.*

Proof. 2^t is positive and increasing in t , and $2 < 3 < 4$. Since $-2^t < 0 < 3$, no signed power of two equals 3. $\square$

Remark 4. The witness 3 is the *smallest* interior entry that is not a signed power of two. The smallest entry that is not a power of two at all is the zero $\begin{bmatrix} 2 \\ 1 \end{bmatrix}_{q=-1} = 0$ (and 0 is not 2^t for any t). Two further small counterexamples are $\begin{bmatrix} 8 \\ 4 \end{bmatrix}_{q=-1} = 6$ and $\begin{bmatrix} 10 \\ 2 \end{bmatrix}_{q=-1} = 5$. For $n \leq 20$ there are 102 nonzero values that are not $\pm$ a power of two.

5 The correct characterisation, and a false one

The true description of the whole table is the $d = 2$ case of the q-Lucas theorem; we state it as the corrected replacement for the conjecture.

Theorem 5 (q-Lucas at $d = 2$; corrected description). *For all $0 \leq k \leq n$,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_{q=-1} = \begin{cases} \begin{pmatrix} \lfloor n/2 \rfloor \\ \lfloor k/2 \rfloor \end{pmatrix}, & \text{if it is not the case that } n \text{ is even and } k \text{ is odd,} \\ 0, & \text{if } n \text{ is even and } k \text{ is odd.} \end{cases}$$

Verification. The rule reproduces the recursion (1) exactly; `reproduce.py` checks it for all $0 \leq k \leq n \leq 20$ with **0 mismatches**, and it visibly explains every entry of the table above (e.g. $\begin{bmatrix} 6 \\ 2 \end{bmatrix}_{q=-1} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} = 3$, $\begin{bmatrix} 8 \\ 4 \end{bmatrix}_{q=-1} = \begin{pmatrix} 4 \\ 2 \end{pmatrix} = 6$, $\begin{bmatrix} 10 \\ 2 \end{bmatrix}_{q=-1} = \begin{pmatrix} 5 \\ 1 \end{pmatrix} = 5$, $\begin{bmatrix} 2 \\ 1 \end{bmatrix}_{q=-1} = 0$ since 2 is even and 1 is odd). This is the q-Lucas congruence specialized to a primitive d -th root of unity at $d = 2$.

Remark 6 (The “parity of $\binom{n}{k}$ ” rule is FALSE). It is tempting to guess

$$\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_{q=-1} \stackrel{?}{=} \begin{cases} \binom{\lfloor n/2 \rfloor}{\lfloor k/2 \rfloor}, & \binom{n}{k} \text{ odd,} \\ 0, & \binom{n}{k} \text{ even.} \end{cases}$$

This is **wrong**, and we explicitly disclaim it. It fails **73 times** for $n \leq 20$. The smallest failure is

$$\left[\begin{smallmatrix} 8 \\ 4 \end{smallmatrix} \right]_{q=-1} = 6, \quad \text{while} \quad \binom{8}{4} = 70 \text{ is even,}$$

so that rule would predict 0 instead of 6. The parity of $\binom{n}{k}$ is irrelevant; the correct criterion in Theorem 5 is the parity of n and of k , not of the binomial coefficient.

Remark 7 (A true but different statement). By Kummer’s theorem the 2-adic valuation of $\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_{q=-1}$ satisfies the true identity

$$v_2 \left(\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_{q=-1} \right) = v_2 \left(\binom{\lfloor n/2 \rfloor}{\lfloor k/2 \rfloor} \right),$$

which is consistent with Theorem 5. This is a statement about the *valuation* of the value, not about the value being a power of two, and it does not rescue the conjecture: e.g. $v_2(3) = 0$, so $2^{v_2(\left[\begin{smallmatrix} 6 \\ 2 \end{smallmatrix} \right]_{q=-1})} = 2^0 = 1 \neq 3$.

6 Alternative readings tried

None of the following rescue the claim.

Reading	Outcome
Value = $2^{v(n,k)}$ (literal, unsigned)	false: $\left[\begin{smallmatrix} 6 \\ 2 \end{smallmatrix} \right]_{q=-1} = 3$ is not a power of two.
Value = $\pm 2^{v(n,k)}$ (signed parenthetical)	false: $3 \neq \pm 2^t$ for every t .
q a primitive d -th root of unity, $d \geq 3$	the values are non-integer algebraic numbers (e.g. $[2 \text{ choose } 1]_\zeta = 1 + \zeta$); not integer powers of two, so the “form 2^v ” claim fails a fortiori.
Restrict to $k = 1$	false: $\left[\begin{smallmatrix} 2 \\ 1 \end{smallmatrix} \right]_{q=-1} = 0$ is not 2^t or $\pm 2^t$ (nor is $\left[\begin{smallmatrix} n \\ 1 \end{smallmatrix} \right]_{q=-1} = 0$ for any even n).
Restrict to even n	false: $\left[\begin{smallmatrix} 8 \\ 4 \end{smallmatrix} \right]_{q=-1} = 6$, an even n , is not a signed power of two.
Read the value as $2^{v_2(\binom{n}{k})}$	false for $n \leq 20$ (the “parity of $\binom{n}{k}$ ” rule) — 73 mismatches, cf. the remark above.
Read v as v_2 of the value itself	false: 3 has $v_2(3) = 0$, so $2^{v_2(3)} = 1 \neq 3$. The true identity $v_2(\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_{q=-1}) = v_2(\binom{\lfloor n/2 \rfloor}{\lfloor k/2 \rfloor})$ concerns valuations only.

Only Theorem 5 correctly describes the value; it is not of the conjectured form, and it produces 3 at (6, 2). The refutation therefore stands under every reading of the filed text.

7 Formalisation in Lean 4

The refutation is formalised in the Lean 4 project under `lean4/` (core Lean only, `import Std`, no `Mathlib`, no `sorry`). The recursion is implemented on integers,

```

def qb : Nat → Nat → Int
| _, 0 => 1
| 0, _+1 => 0
| n+1, k+1 => qb n k + (-1 : Int)^(k+1) * qb n (k+1)

```

so that `decide` can evaluate it (structural recursion, no division, no `Rat`). The main statements are

```

theorem qb_6_2 : qb 6 2 = 3
theorem qb_8_4 : qb 8 4 = 6
theorem pow2_ne_three : ∀ t : Nat, (2 : Int)^t ≠ 3
theorem negpow2_ne_three : ∀ t : Nat, -((2 : Int)^t) ≠ 3
theorem not_pow2_unsigned : ¬ (∀ n k, ∃ t, qb n k = 2^t)
theorem not_pow2_signed : ¬ (∀ n k, ∃ t,
  qb n k = 2^t ∨ qb n k = -(2^t))
theorem conjecture_00000000429_false :
  ¬ (∃ v : Nat → Nat → Nat, ∀ n k, qb n k = (2 : Int)^(v n k))

```

`Check.lean` prints `#print axioms` for every theorem and reports only `propext` and `Quot.sound`; in particular there is no `sorryAx`. The concrete evaluations (`qb_6_2` etc.) depend on no axioms at all. See `lean4/README.md` for the full statement table.

8 Reproducibility

```

$ python3 reproduce.py
... (exact integer Pascal recursion, independent polynomial cross-check,
the q-Lucas rule, the false parity rule, the 102-value census)
PASS: all checks verified; conjecture 00000000429 is FALSE.
$ tectonic --outdir build main.tex
$ cd lean4 && lake build && lake env lean Check.lean

```

`reproduce.py` uses the Python 3 standard library only and exits non-zero if any check fails. The value $\left[\begin{smallmatrix} 6 \\ 2 \end{smallmatrix} \right]_{q=-1} = 3$ is confirmed three ways: by the integer recursion (1), by exact polynomial arithmetic (the Gaussian binomial divided out as a polynomial in q and then evaluated at $q = -1$), and in Lean by `decide`.

Status against the submission rules

Rule 3 requires a LaTeX source, a PDF document, and a Lean 4 project.

- **LaTeX source** — `main.tex`, a standalone article using `geometry`, `amsmath`, `amssymb`, `amsthm`, `array`, `parskip`, `booktabs`; compiles with `tectonic main.tex`.
- **PDF** — at `build/main.pdf`, produced by `tectonic --outdir build main.tex`.
- **Lean 4 project** — `lean4/` with `lean-toolchain` pinned to `leanprover/lean4:v4.33.1`, `lakefile.toml` (library `tlmc429`, no dependencies), `Main.lean`, `Check.lean` (`#print axioms`), and `lean4/README.md`. `lake build` exits 0 and the audit reports no `sorryAx`.